Saman Abeysinghe

AI Engineer

saman.amk@gmail.com · Sri Lanka
linkedin.com/in/saman-abeysinghe-6460a0174 · github.com/samanAbeysinghe

SUMMARY

AI Engineer with a BSc (Hons) in Engineering, specialised in Computer Engineering. I design and build production-ready AI applications using LLMs, RAG, multi-agent architectures, document intelligence and computer vision. Currently at Cenango (Pvt) Ltd, with earlier experience in backend development, machine learning and embedded systems.

EXPERIENCE

AI Engineer
2025 – Present

Cenango (Pvt) Ltd

- Developing AI-powered systems built on LLMs, RAG, and multi-agent architectures.

Temporary Demonstrator
2025

The Open University of Sri Lanka

- Conducted laboratory sessions for Microprocessors & Interfacing, Object Oriented Programming, Software Development for Engineers, and Computer Technology.

Visiting Technician
2024

The Open University of Sri Lanka

- Conducted laboratory sessions for the Microprocessors & Interfacing course.
- Worked with micro-controller programming and simulation software.

Software Engineer Trainee
2023 – 2024

The Open University of Sri Lanka

- Implemented backend services using Spring Boot.
- Managed and maintained a PostgreSQL database.
- Wrote C++ code to program micro-controllers.

Research Engineer Trainee
2022

NERDC

- Implemented C++ programs for micro-controller based projects.
- Used simulation software to run and validate simulations.

SELECTED AI PROJECTS

RAG-Based Chatbot for WordPress Sites

Claude · Voyage AI · Qdrant · Flask · AWS Bedrock · SQS/SNS · PostgreSQL

- Built end to end: a self-serve multi-tenant platform where customers register on a dashboard and connect their WordPress site.
- The system crawls and indexes the site content, then issues a keyed script tag that embeds a Claude-powered chat widget answering visitor questions from that content.
- Content ingestion runs asynchronously through AWS SQS/SNS.

Document & Handwriting Extraction System

YOLO · Claude (AWS Bedrock) · Next.js · Flask · PostgreSQL

- Reads unstructured documents containing handwritten English text, detecting question and answer fields separately and returning structured question-and-answer data.
- Users can edit, search, delete and add extracted fields, and view the fully annotated document.
- A rule engine evaluates each document against user-defined rules and reports which passed and failed.

Fully Local, Private RAG System

Ollama · ChromaDB · React · Flask · PostgreSQL

- Chat system that runs entirely on the user's machine with no internet access, for sectors such as banking and legal work that need a private environment.
- Users upload any documents and query across all of them, manage the document inventory, and rate each response so the system keeps improving.

QA Copilot: AI-Assisted Software Testing (MSc research project)

Next.js · TypeScript · Flask · SQLite · Ollama (Llama 3.1 8B)

- Built solo: a full-stack platform (77 API endpoints, 23 tables, 26 automated tests) where a local LLM drafts test cases from user stories, people approve every one, and sprints run with AI on or off to measure whether it helps.
- Schema-validated output, versioned prompts, and embedding-based duplicate detection calibrated on measured scores. Based on a review of 15 existing AI testing tools; the research survey is still to be run.

MACHINE LEARNING PROJECTS

Nutrient Deficiency Detection for Black Pepper Leaf

TensorFlow · Python · Flask · HTML/CSS/JavaScript

- Web app that checks whether an uploaded image is a black pepper leaf, identifies the nutrient deficiency, and recommends how to mitigate it. Published at SLAAI-ICAI 2024 (see Publications).

Student Dropout Prediction System

TensorFlow · Python · Flask · HTML/CSS/JavaScript

- Predicts whether a newly registered student will graduate, drop out, or stay enrolled, based on registration data entered through a web interface.

AI-Powered Health Symptom Checker

TensorFlow · Python · Flask · HTML/CSS/JavaScript

- Splits symptoms entered by a patient into keywords, predicts the likely disease with a pre-trained ML model, then scrapes the web to present details and medication information.

Other projects: IoT real-time power monitoring system (Spring Boot, Angular, PostgreSQL, C++); wood classification and defect detection (React, Flask, YOLO, PyTorch); recipe management website (React, Spring Boot, OAuth2, JWT, MySQL); automated public space management system for public transport (Pera Innovations, University of Peradeniya); matrix operations processor (VHDL, FPGA).

TECHNICAL SKILLS

AI & LLMs	Claude, AWS Bedrock, Voyage AI, Llama 3.1, RAG, multi-agent systems, Ollama, Qdrant, ChromaDB
Document AI & Vision	YOLO, TensorFlow, PyTorch, OCR and handwriting extraction
Languages	Python, TypeScript/JavaScript, Java, C++, C, HTML, CSS
Backend & Data	Flask, Django, Spring Boot, PostgreSQL, MySQL, SQLite, AWS SQS/SNS
Frontend	React, Next.js, Angular
Embedded	C++ micro-controllers, VHDL, FPGA, simulation tools

EDUCATION

BSc (Hons) in Engineering, specialised in Computer Engineering – Second Class Upper Division 2018 – 2024
The Open University of Sri Lanka

GCE Advanced Level – Mathematics stream (3 passes) 2017
St. Sylvester's College, Kandy

Following AWS training and certification courses.

PUBLICATION

A. M. S. K. Abeysinghe and G. Anthonys, *"Nutrient Deficiency Detection System for Black Pepper Leaf Using Machine Learning,"* 2024 8th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI). DOI: 10.1109/SLAAI-ICAI63667.2024.10844967.

AWARDS

Dean's List of Distinguished Students, 2020/2021

LANGUAGES

English, Sinhala

VOLUNTEERING

Students for the Exploration and Development of Space (SEDS) Sri Lanka – Director, IT division

Zero Plastic Planet – Member

REFERENCES

Eng. Dr. G. Anthonys
Senior Lecturer, Department of Electrical and Computer Engineering, The Open University of Sri Lanka, Nawala, Nugegoda
ganth@ou.ac.lk

Eng. W.D.S.S. Bandara
Senior Lecturer, Department of Electrical and Computer Engineering, The Open University of Sri Lanka, Nawala, Nugegoda
wdban@ou.ac.lk